

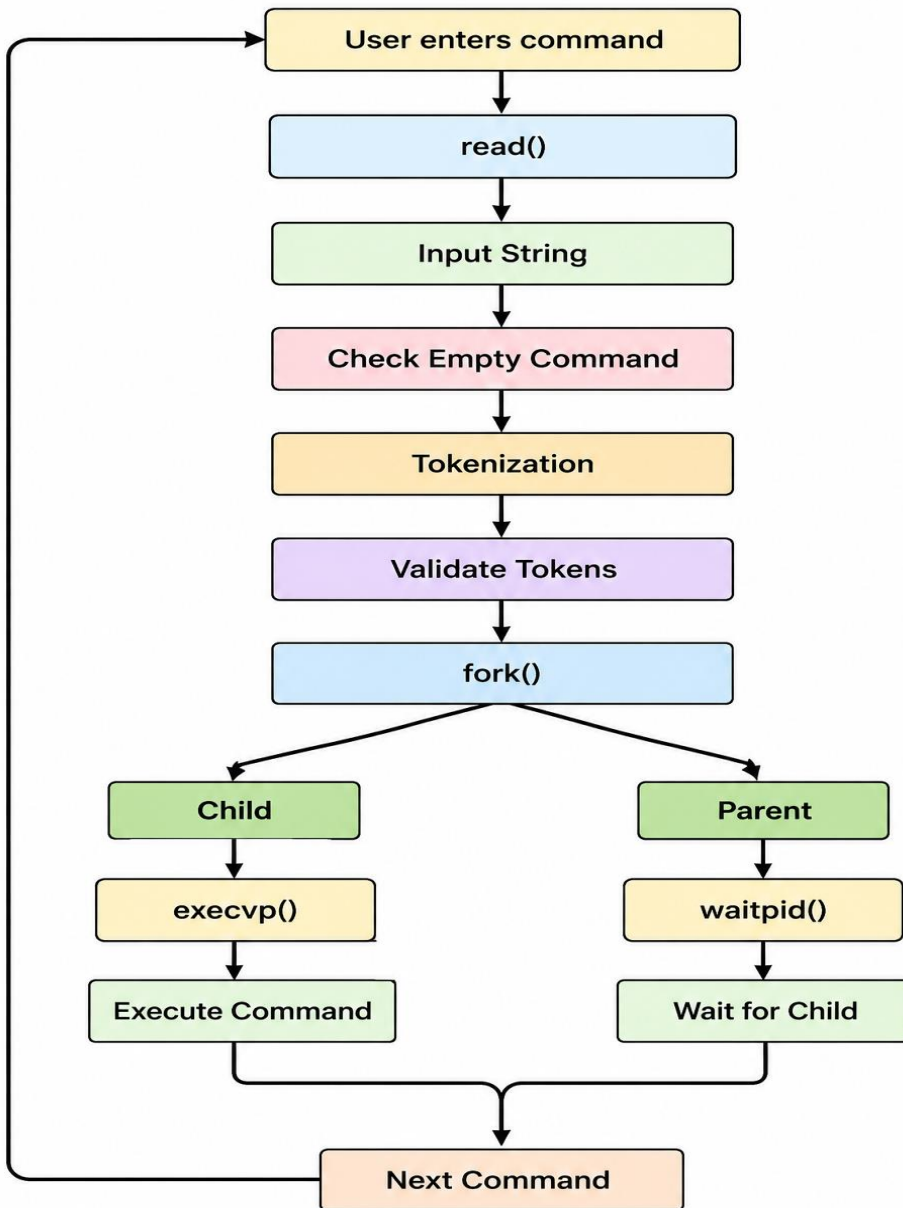
Refer the following link to understand the Parsing and Lexical analysis
<https://www.geeksforgeeks.org/compiler-design/introduction-of-lexical-analysis/>

Task 1:

```
sathwika@123:~$ ./myshell
myShell> echo Hello
Parsing Result:
Argument 1: echo
Argument 2: Hello
Child PID: 3210
Hello
Command execution completed.
```

To Split Input into Tokens, Identify Delimiters, Handle Whitespace, Create Token Structures, Validate Token Streams, Debug Parsing Output and Design Parser Logic, Generate Parse Trees, Validate Syntax, Detect Errors, Handle Empty Commands, Produce Execution Structures

Flow Chart:

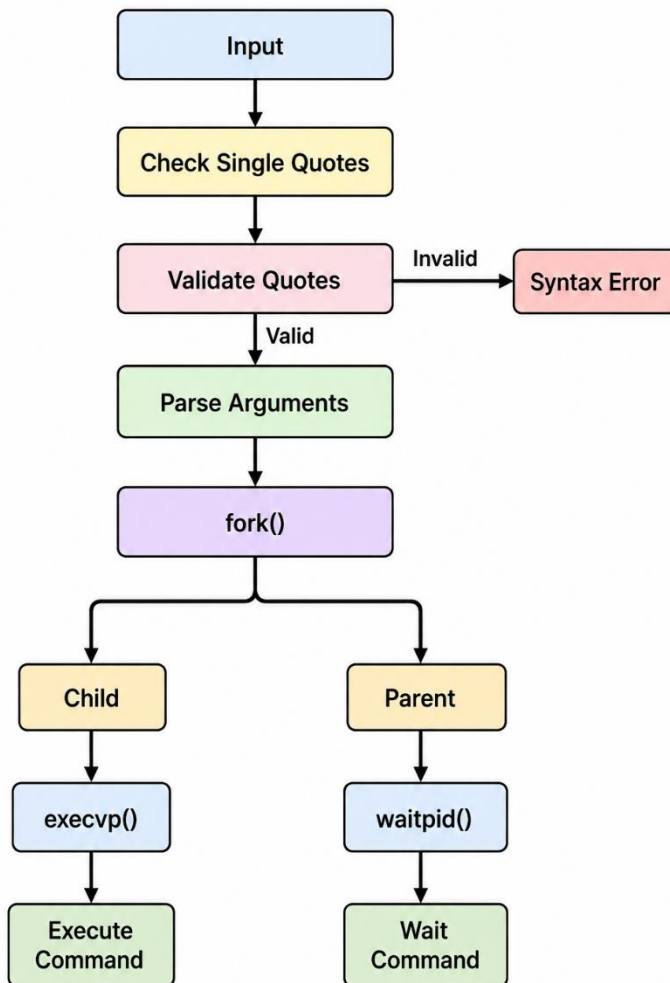


Task 2:

```
sathwika@123:~$ ./myshell
myShell> echo Hello
Parsing Result:
Argument 1: echo
Argument 2: Hello
Child PID: 3210
Hello
Command execution completed.
```

To Apply Single Quotes, Preserve Literal Content, Ignore Variable Expansion, Store Quoted Strings, Validate Parsing Results, Test Edge Cases.

Flow Chart:



Task 3:

```
sathwika@123:~$ ./myshell
myShell> echo "Hello $USER"
--- Parsed Arguments ---
Argument 1: echo
Argument 2: Hello raghu
Child Process
PID = 3022
Hello raghu
Child process completed.
```

To Apply Double Quotes, Preserve Spaces, Allow Variable Expansion, Parse Nested Tokens, Validate Outputs, Test Quoted Commands.

--- Parsed Arguments ---

Child process completed.

Task 4: To Create Process Escape Sequences, Handle Escaped Spaces, Escape Special Symbols, Preserve Characters, Validate Parser Output, Test Complex Inputs.

```
sathwika@123:~$ ./myshell
myShell> echo Hello\ World
Argument 1: echo
Argument 2: Hello World
Hello World
Command execution completed.
```

Task 5: To Create Child Processes, Execute Programs, Handle Execution Errors, Pass Arguments, Manage Parent Process, Test Command Launching.

```
sathwika@123:~$ ./myshell  
myshell$ ls  
Parent Process PID: 1200  
Created Child PID: 1201  
Child Process PID: 1201  
file1.txt  
file2.txt  
Child exited with status: 0
```

Flow Chart:

